

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Examination Summer 2024****Course: B. Tech****Branch: Civil & Infra/ Civil Engineering****Semester: V****Subject Name: Design of Steel Structures****Subject Code: BTCIC501/ BTCVC501****Max Marks: 60****Date: 01/07/2024****Duration: 3 Hrs.****Instructions to the Students:**

1. All questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mentioned it clearly.
5. Use IS 800, Steel table and IS 875 are permitted

		Level/CO	Marks
Q. 1	Solve any TWO of the following		12
(A)	State the advantages and disadvantages of steel structures	Understand	06
(B)	Calculate the strength of ISA 90X60X8 mm used as a tie member with its longer leg connected by 16mm diameter rivets.	Analyze	06
(C)	Compare rivet and welding connections?	Synthesis	06
Q.2	Solve any TWO of the following		12
(A)	An angle of ISA 50X50X6 mm is to carry an axial load of 72KN. The angle is to be connected to 8mm thick gusset plate. Design welded connection.	Application	06
(B)	Design a riveted connection for single ISA 75 x 75 x 8mm connected to 10mm thick gusset plate by using 10mm dia. Rivets at site to carry a load of 90KN.	Application	06
(C)	Distinguish between web buckling and web crippling?	Synthesis	06
Q. 3	Solve any TWO of the following		12
(A)	Describe the components of gantry girder with the help of neat sketch.	Knowledge	06
(B)	Calculate dead load per panel for a roof truss using following details: a) Span of truss = 20m b) spacing = 4m c) rise = 3m d) Number of panels = 10 m e) weight of covering = 100N/ M ² f) Self weight of truss = 105 N/ M ² g) weight of bracing = 75N/ M ² h) weight of covering = 150N/ M ²	Application	06
(C)	Describe the component part of industrial truss.	Knowledge	06
Q. 4	Solve any TWO of the following		12
(A)	Design a column using I- section of length 5m subjected to an axial load of 1175 KN. The column is effectively held in position at both ends and not restrained against rotation at one end. Take $f_y = 250\text{Mpa}$.	Application	06
(B)	Write short note on Gusseted base foundation.	Synthesis	06
(C)	Design a slab base for a column section consists of two channels ISMC 400 placed back to back with 200mm gap. The laced column carry load of 1350 KN. Use M15 Grade concrete for pedestal.	Application	06
Q. 5	Solve any TWO of the following		12
(A)	Explain idealized stress-strain curve for mild steel.	Knowledge	06
(B)	Distinguish between plastic design and elastic design.	Synthesis	06

(C)	Explain the concept of shape factor and load factor	Synthesis	06
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